



Data Analytics Project: IGN Game Analysis

Dataset Used: ign

This document outlines a structured, end-to-end data analysis project based on IGN game reviews data. The project is divided into 5 scenarios like Data Preprocessing, data handling, Numpy operations, analysis and visualization.

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Dataset: Ign.csv (A csv file format)

Project Goal: To analyze historical trends, platform performance, genre and scoring patterns across thousands of game reviews published by IGN.

Modules Used:

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth using `np.diff()`.
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts and Histogram).

Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff())
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart, Histogram.

Scenario 1: Data Loading & Preprocessing (Easy)

Description:

This scenario focuses on importing ign game reviews data from a CSV file into a Pandas Data Frame. Missing values are spotted and fixed to keep the data clean and consistent. After that, the necessary columns are converted into numeric data types to perform accurate analysis and data visualization.

Task	Description
1.	Load the dataset(ign.csv) into a Pandas DataFrame (pd.read_csv()).
2.	Display the first and last 5 rows and shape of dataset to understand the structure of Dataset.
3.	Removed the unnecessary column 'Unnamed :0' (index column)
4.	Check for missing values in score, genre and platform and replace missing values in score with mean and genre with mode.
5.	Convert score feature to float datatype and release_year, release_month, release_day to int datatype.

Result:

```
First 5 rows from the dataset:
  Unnamed: 0 score_phrase title ... release_year release_month release_day
0          0    Amazing LittleBigPlanet PS Vita ...      2012           9         12
1          1    Amazing LittleBigPlanet PS Vita -- Marvel Super Hero E... ...      2012           9         12
2          2     Great Splice: Tree of Life ...      2012           9         12
3          3     Great NHL 13 ...      2012           9         11
4          4     Great NHL 13 ...      2012           9         11
```

```
Last 5 rows from the dataset:
  Unnamed: 0 score_phrase title ... release_year release_month release_day
18620      18620     Good Tokyo Mirage Sessions #FE ...      2016           6         29
18621      18621    Amazing LEGO Star Wars: The Force Awakens ...      2016           6         29
18622      18622   Mediocre Star Ocean: Integrity and Faithlessness ...      2016           6         28
18623      18623 Masterpiece Inside ...      2016           6         28
18624      18624 Masterpiece Inside ...      2016           6         28

[5 rows x 11 columns]
```

```
Shape of Dataset: (18625, 11)
```

Conclusion:

The dataset was successfully loaded and prepared for analysis by handling missing values and ensuring proper data types. Missing values in score have been replaced with mean and missing genre entries are filled with mostly repeated categorical data (mode). Unnecessary column named 'Unnamed: 0' have been removed from the dataset to reduce noise. This cleaning process improves data consistency and reliability. After following these preprocessing steps, the dataset is now well structured and ready for deeper analysis in the subsequent scenarios.

Scenario 2: Line Graph - Score trend(🌀 Easy)

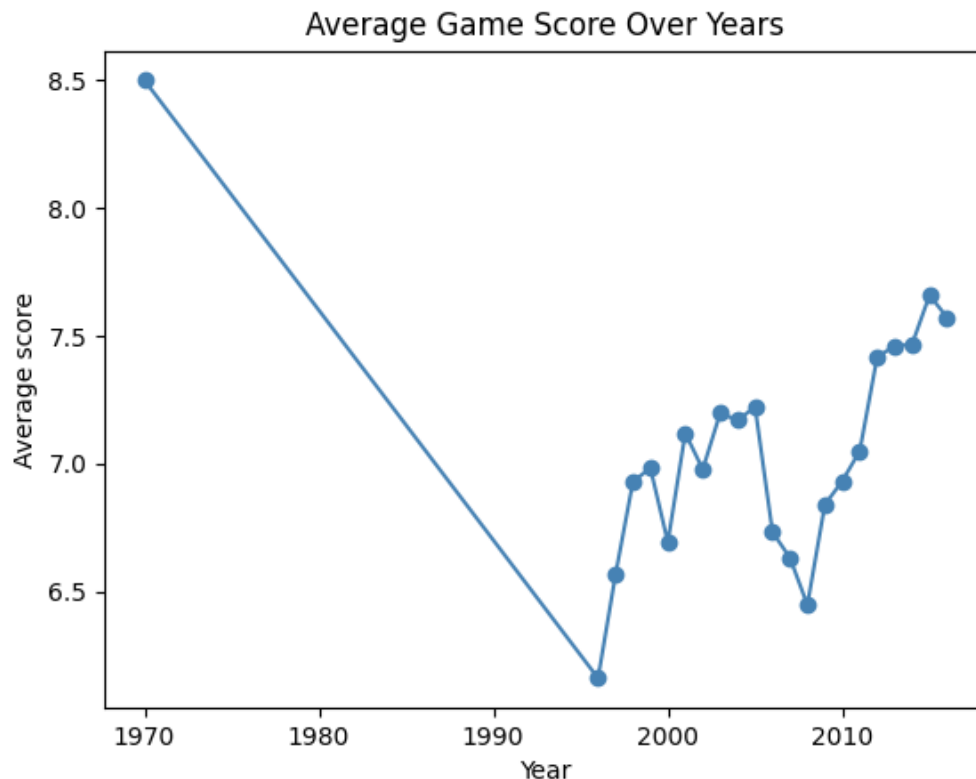
Description:

In this scenario, data is grouped by release year and average score per year calculated using pandas. Later this resulting data is converted to a numpy array for numerical operations. A line graph is plotted between average score and release year which shows the average score changed over years.

Task	Description
1.	Group data by release year using <code>groupby()</code> .
2.	Calculate average score per year using <code>mean()</code> .
3.	Covert result into the numpy array to use in data visualization.
4.	Plot a line graph between <code>release_year</code> on x-axis and average score on y-axis.
5.	Add title ("Average Game Score Over Years") and proper labels (<code>xlabel</code> , <code>ylabel</code>) to the line graph.
6.	Save the graph into the system using <code>savefig()</code> .

Graph:

A line graph titled “Average Game Score Over Years” is plotted with `release_year` on x-axis and corresponding average score on y-axis. This shows whether game review standards improved, declined or remain stable over the years.



Conclusion:

The line graph of IGN game scores reveals a significant decline from 8.5 in 1970 to a low of 6.2 in the mid-1990s, followed by a steady recovery reaching 7.5–7.7 by the mid-2010s. Every peak and dip reflects real shifts in game quality and review standards over time. This analysis also effectively demonstrates the use of Pandas grouping, NumPy array manipulation, and proper graph labelling as essential tools for analysing and visualizing trends in data.

Scenario 3: Filtering + Bar Chart (🌀 Medium)

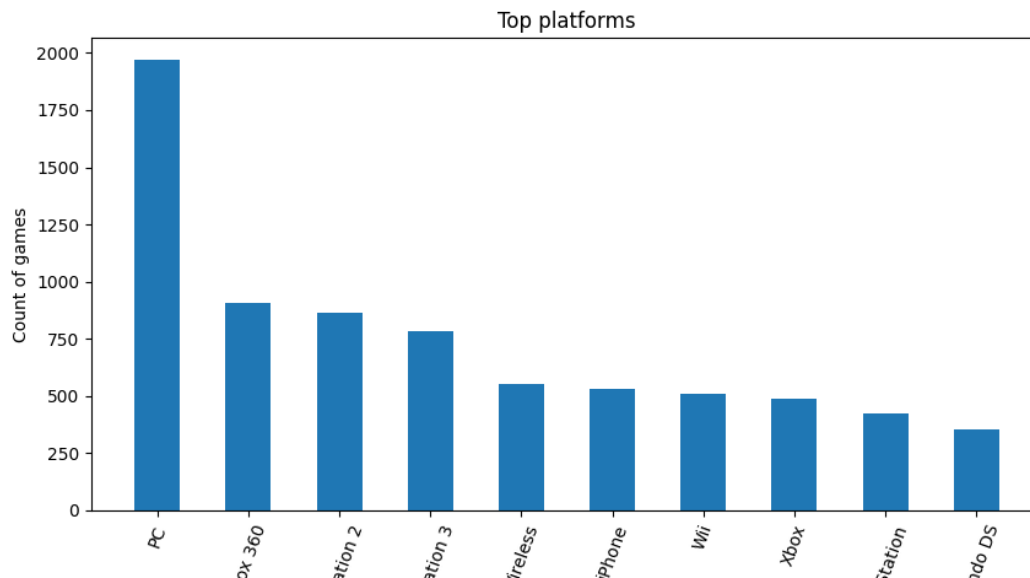
Description:

This scenario focuses on filtering the dataset by taking only the high rated games per platform, where score of a game is greater than 7. Then a bar chart is plotted to compare the top 10 platforms using platform as x-axis label and count of games as y-axis label. This helps in creating a clear comparative visualization for categorical data.

Task	Description
1.	Filter the dataset to consider only the rows where score is greater than 7 (High rated games).
2.	Count the number of High rated games per platform using <code>groupby()</code> and <code>count()</code> .
3.	Select top 10 platforms form the data using <code>head(10)</code> .
4.	Convert data into numpy arrays, that will be suitable for plotting.
5.	Plot a bar graph using platform on x-axis and count of games on y-axis.
6.	Rotate the x-axis label to increase the readability using <code>xticks(rotation=70)</code> .

Graph:

A vertical bar chart titled “Top 10 Platforms” is plotted with platform names on x-axis (used rotation to display data clearly) and number of high rated games on y-axis. This graph visually shows the category wise high rated games using the bar graph to clearly understand the data.



Conclusion:

The bar chart clearly shows the number of high-rated games across the top 10 platforms based on IGN review scores. PC stands out as the leading platform, with nearly 2,000 highly rated games, far ahead of the others. While other platforms such as Xbox 360 and PlayStation 2 PlayStation 3, Wii, Xbox, iPhone, and Nintendo DS follow up the remaining positions. This indicates that PC has the strongest presence in terms of well-reviewed game titles, likely due to its larger game library and broader genre variety. Overall, the chart effectively highlights platform performance and demonstrates how Pandas filtering and categorical visualization can be used to draw meaningful insights from the dataset.

Scenario 4: Aggregation+ Pie Chart (🌀 Medium)

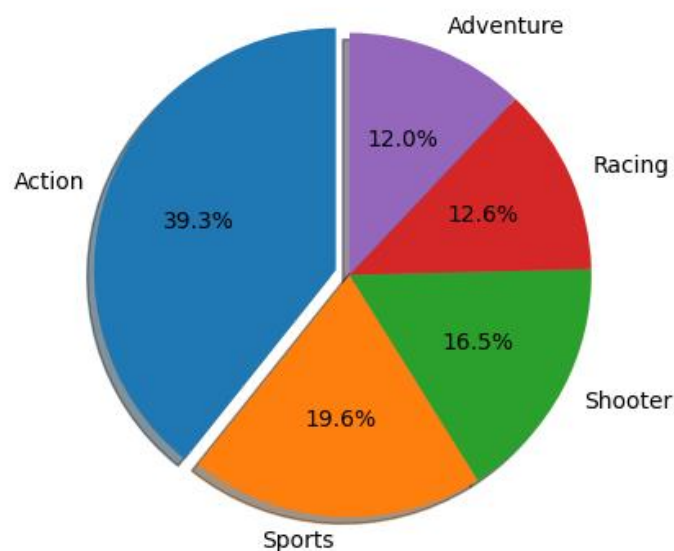
Description:

This section focuses on analysing how games are spread across different genres. Using Pandas, the dataset is grouped by genre and the total number of games in each category. Top 5 genres are picked and used for visualization through a pie chart, where each slice represents a genre and its percentage share, making it simple to see which genres dominate the gaming world.

Task	Description
1.	Count the number of games per genre.
2.	Top 5 genres are selected using head().
3.	Prepare labels (genre names) and values (game counts).
4.	Plot a pie chart using Labels (genre) and Values (counts).
5.	Add percentage labels to each slice using autopct='%1.1f%%'. Save the graph using plt.savefig()

Graph:

A pie chart is plotted to visualize the share of most common genres in the IGN dataset, with each slice labelled by genre name and percentage labels shown using autopct. The results clearly show that Action dominates at 39%, nearly double that of Sports at 19.7%, while Shooter, Racing, and Adventure hold balanced shares of 16.6%, 12.6%, and 12.1% respectively, confirming that Action is the most dominant genre while the remaining four share a fairly even distribution.



Conclusion:

The IGN dataset is grouped by genre and used top 5 observations for visualization using pie chart. Action games dominate by a wide margin at 39%, nearly doubling the second place Sports genre at 19.7%, while Shooter, Racing, and Adventure follow closely with balanced shares. This analysis makes it evident that Action is the most produced and well-received genre, while the remaining genres still hold a fair and competitive presence in the gaming industry.

Scenario 5: Advanced Analysis + Multiple Graphs (🟡 Hard)

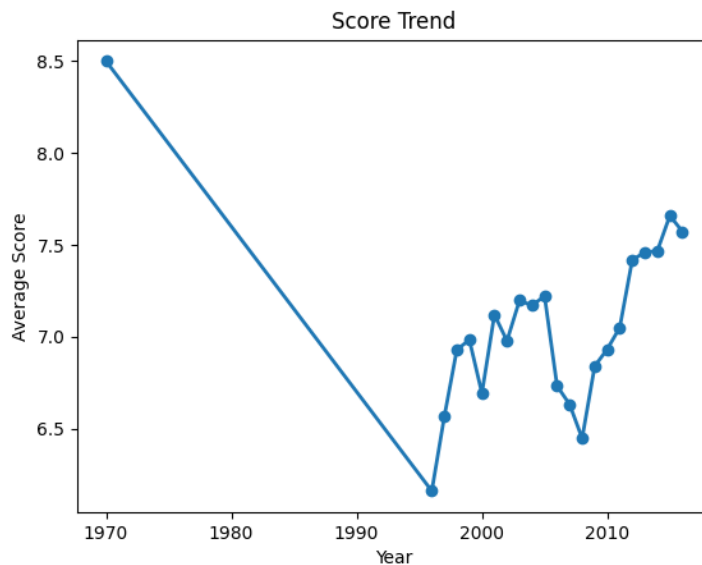
Description:

This scenario is the comprehensive part of the project, combining feature engineering, NumPy-based numerical analysis, and multiple visualizations all in one place. Game scores are categorized into meaningful labels, a binary column is created for Editors' Choice awards, and used Numpy to calculate yearly score growth. A line graph is plotted to analyse scoring trends , a stacked bar chart to analyse category distribution over time, and a histogram to uncover insights about the relationship between Editors' Choice awards and high scores, making this the most well-rounded and insightful scenario in the entire project.

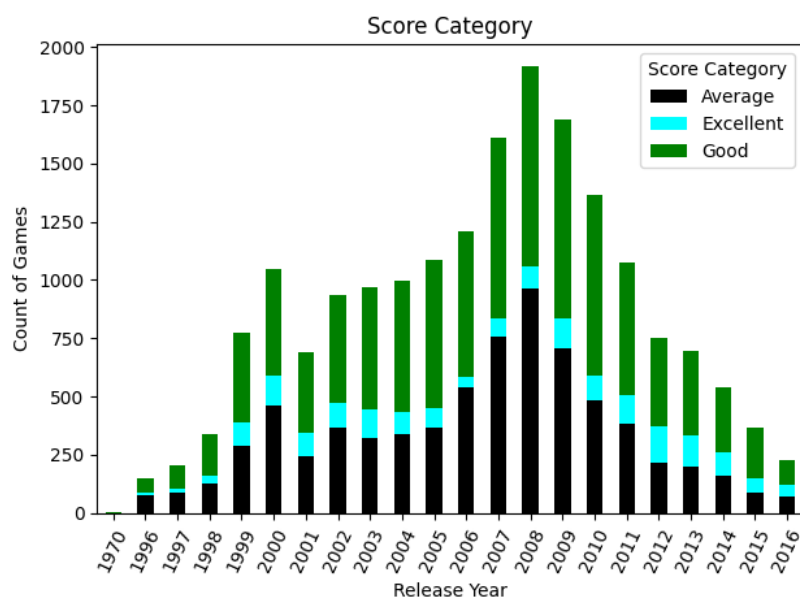
Task	Description
1.	Create a new column named <code>score_category</code> to categorize score into, “Excellent” for <code>score >=9</code> , “Good” for <code>7<=score<9</code> and “Average” for <code>score<7</code> .
2.	Convert the Editors choice <code>Y</code> →1 and <code>N</code> →0.
3.	Calculate Yearly growth using <code>np.diff()</code> based on the average yearly scores calculated before.
4.	Line Graph: Plot a graph showing the average score per year to observe scoring trends.
5.	Stacked bar chart: Plot a graph to show count of each <code>score_category</code> (Excellent, Good, Average) per <code>release_year</code> to visualise shifts in review quality over time
6.	Histogram: Plot the distribution of Score
7.	Save all the graphs using <code>plt.savefig()</code> <code>plt.savefig("score_trend.png")</code> <code>plt.savefig("score_category_stacked.png")</code> <code>plt.savefig("score_distribution.png")</code>
8.	Identify which years recorded the highest average review scores, marking the peak quality periods in the dataset. Also determines whether Editors' Choice games consistently correlate with higher scores validating the editor’s choice process.

Graph:

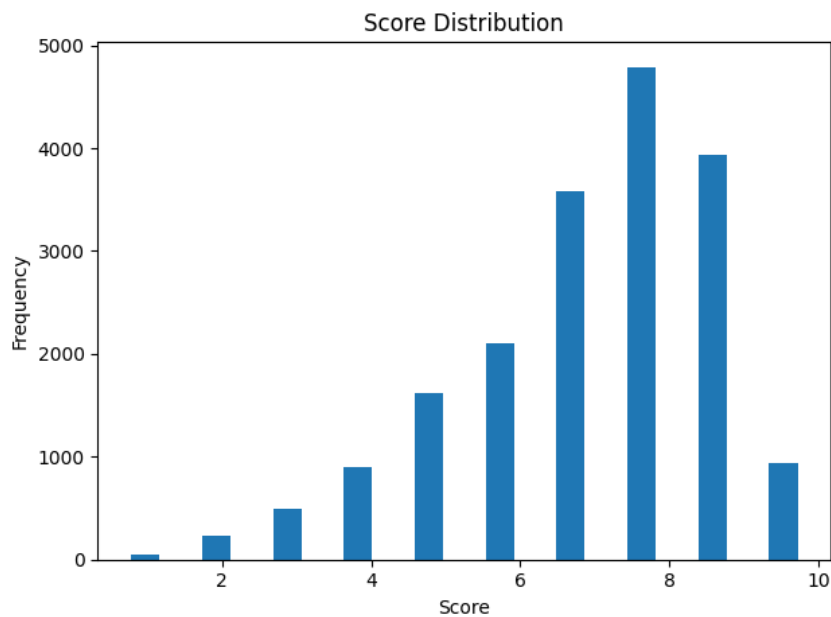
Line Graph: This graph shows the average game scores from 1970 to 2017. Scores started high at 8.5 in 1970, dropped sharply to a low of 6.2 in the mid-1990s, and then gradually recovered back to around 7.5–7.7 by 2017, indicating an overall improvement in modern game quality.



Stacked bar chart: The stacked bar chart displays the count of games across three score categories: Excellent, Good, and Average, for each release year from 1970 to 2016. Game releases peaked around 2008–2009 with nearly 2000 titles, with the Good category dominating throughout most years, while Excellent rated games remained consistently small across all years.



Histogram: The graph shows how game scores are distributed across the entire dataset. The majority of games are clustered around scores of 7 to 9, with the highest frequency near score 8, suggesting that most games in the IGN dataset tend to receive above average to good ratings overall.



Conclusion:

This scenario brings everything together by combining feature engineering, NumPy analysis, and three distinct visualizations into one complete picture. The `score_category` feature enables categorical trend analysis, while `np.diff()` measures yearly growth. The line graph, stacked bar chart, and histogram together cover trends, composition, and distribution simultaneously, offering a truly multi-dimensional view of the dataset. Ultimately, this scenario provides a holistic understanding of how game quality has evolved over time through IGN's lens and whether the Editors' Choice designation genuinely aligns with higher scoring games.

Final Conclusion:

This project demonstrates the use of Pandas, NumPy, and Matplotlib in analyzing and visualizing a real-world gaming dataset across multiple scenarios. The findings revealed that PC dominates in high-rated game count, Action is the most popular genre, and average game scores showed a long-term recovery after a significant decline in the mid-1990s. Overall, this project highlighted how structured data analysis and visualization can effectively transform raw data into clear and actionable insights, proving that the right tools and techniques can bring any dataset to life in a meaningful way.

What you have learned

Through this project, I developed a strong understanding of how Pandas can be used for data filtering, grouping, aggregation, and creating meaningful column transformations like score categories. I also gained a deeper appreciation for Numpy's ability to handle numerical data efficiently, especially using `np.diff()` to track yearly score changes. Building different chart types from line graphs to stacked bar charts and histograms gave me confidence in choosing the right visualization for the right story. Most importantly, this project taught me how to seamlessly connect data preparation, analysis, and visualization into a complete end-to-end workflow that turns raw, unstructured data into clear and actionable conclusions.